Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	PMAT1/N01

Past Paper 1

Time: 25 minutes | Paper Reference **PMAT1/N01**

Mathematics

Level 1
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks: 14 /14

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and your answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**
- Take the value of π to be 3.14

Information

- The total mark for this section is 14.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1

(a) Convert 3 metres into centimetres.

(1)

Step 1: Metric conversion factor (1 m = 100 cm):

- *1 metre = 100 centimetres*
- **3 metres = 3 × 100 = 300 cm [A1]**

300 *[A1]* cm

(b) Work out 3400 ÷ 1000

(1)

Step 1: Dividing by 1000 shifts digits 3 places right:

- *3400 ÷ 10 = 340*
- *340 ÷ 10 = 34*
- **34 ÷ 10 = 3.4 [B1]**

3.4 *[B1]*

(c) Calculate 35 – 15 ÷ 5

(1)

Follow BIDMAS: Division must be done before Subtraction:

- *Step 1: 15 ÷ 5 = 3*
- **Step 2: 35 – 3 = 32 [C1]**

32 *[C1]*

(Total for Question 1 is 3 marks)

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2 Aaron is building a brick wall.

The wall will need 1025 bricks.
The weight of each brick is 2.89 kg.

Aaron will pay a delivery charge if the total weight of the bricks is over 2500 kg.

Use estimation to work out if you think Aaron will need to pay the delivery charge.
You **must** show your working.

(3)

Step 1: Round values to 1 significant figure / manageable numbers:

- *1025 bricks rounds to 1000 bricks [A1]*
- *2.89 kg rounds to 3 kg [A1]*

Step 2: Estimate the total weight of the bricks:

Estimated weight = 1000 × 3 kg = 3000 kg [B1]

Step 3: Compare estimated weight with delivery threshold:

- *Delivery threshold = 2500 kg*
- *3000 kg > 2500 kg (weight exceeds 2500 kg threshold)*

Conclusion: Yes, Aaron will need to pay the charge! [C1]

Alternative division method check:

- *2500 kg ÷ 3 kg ≈ 833 bricks*
- *1025 bricks > 833 bricks --> Weight exceeds 2500 kg*

Yes [C1]

(Total for Question 2 is 3 marks)

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3 A farmer has a rectangular field.
The field is 300 m long and 200 m wide.

(a) Calculate the perimeter of the field.

(2)

Step 1: Calculate perimeter (sum of all 4 rectangular sides):

Perimeter = 300 + 200 + 300 + 200 [M1]

= (2 × 300) + (2 × 200) = 600 + 400

= 1000 m [A1]

1000 [A1] m

The farmer is buying fence panels.
He needs a total length of 200 m of fence panels.

Each fence panel is 2.5 m in length.

(b) Work out how many fence panels the farmer will need to buy.

(2)

Step 1: Divide total length by length of one fence panel:

Number of panels = 200 ÷ 2.5 [M1]

Step 2: Mental calculation (double both values):

- 200 ÷ 2.5 = 400 ÷ 5 = 80 fence panels [A1]
- Check: 80 × 2.5 m = 200 m (matches required length)

80 [A1]

(Total for Question 3 is 4 marks)

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4 Saira is buying a new car.
She will keep the car for 5 years.

Saira has this information.

Diesel car	**Petrol car**
Buying price £29 515	Buying price £26 775
Average running cost £1200 each year	Average total running cost £8000 for 5 years

Saira wants to see whether a diesel car or a petrol car is better value for the 5 years.

Which type of car is better value?
Show why you think this.

(4)

Step 1: Calculate 5-year running cost for the diesel car:

5 × £1200 per year = £6000 [M1]

Step 2: Calculate total cost over 5 years for each car:

- *Petrol car: £26 775 + £8000 = £34 775 [M1]*
- *Diesel car: £29 515 + £6000 = £35 515 [M1]*

Step 3: Compare both total 5-year costs:

- *Difference = £35 515 - £34 775 = £740*
- *£34 775 < £35 515 (Petrol car is £740 cheaper)*

Conclusion: The Petrol car is better value! [A1]

Petrol car [A1]

(Total for Question 4 is 4 marks)

TOTAL FOR SECTION A IS 14 MARKS

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Centre Number	Candidate Number
MFS-UK	PMAT1/C01

Past Paper 1

Time: 1 hour 30 minutes | Paper Reference **PMAT1/C01**

Mathematics

Level 1
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks: 42 /42

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- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and your answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used.**
- If your calculator does not have π button take the value of π to be 3.14

Information

- The total mark for this section is 42.
- The total mark for this paper is 56.
- The marks for each question are shown in brackets
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Advice

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- Check your answers if you have time at the end.

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Johan works in a cafe.
One morning he sells 48 cups of tea and 12 cups of coffee.

(a) Write down the ratio of the number of cups of tea to the number of cups of coffee.
Write your answer in its simplest form.

(2)

Step 1: Write initial ratio of tea to coffee:

Tea : Coffee = 48 : 12 [M1]

Step 2: Simplify by dividing both parts by 12 (HCF):

48 ÷ 12 = 4 and 12 ÷ 12 = 1

Simplest form = 4 : 1 [A1]

4 : 1 [A1]

A customer buys four cups of tea.
Johan puts sugar in two of the cups of tea.
The customer picks one cup at random.

(b) What is the likelihood that the customer picks a cup of tea with sugar?

(1)

Tick [✓] a box to show your answer.

[ ] impossible

[ ] unlikely

[✓] even chance 2 of 4 cups = 1/2 (50%) --> Even chance [B1]

[ ] likely

[ ] certain

(Total for Question 1 is 3 marks)

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2 Sam is a manager in an office.
She wants to buy 14 new chairs.

Each chair costs £73
Sam will also be charged 20% VAT on the cost of the chairs.

Sam has £1200 to buy the chairs.

Is £1200 enough to buy the 14 chairs?

(4)

Step 1: Calculate total net cost of 14 chairs (£73 each):

14 × £73 = £1022 [M1]

Step 2: Calculate 20% VAT on the chairs:

20% of £1022 = 0.20 × 1022 = £204.40 [M1]

- *10% of £1022 = £102.20*
- *20% = £102.20 × 2 = £204.40*

Step 3: Calculate total cost including VAT:

Total gross cost = £1022 + £204.40 = £1226.40 [M1]

Step 4: Compare with Sam's budget of £1200:

- *£1226.40 > £1200 (Sam is £26.40 short)*

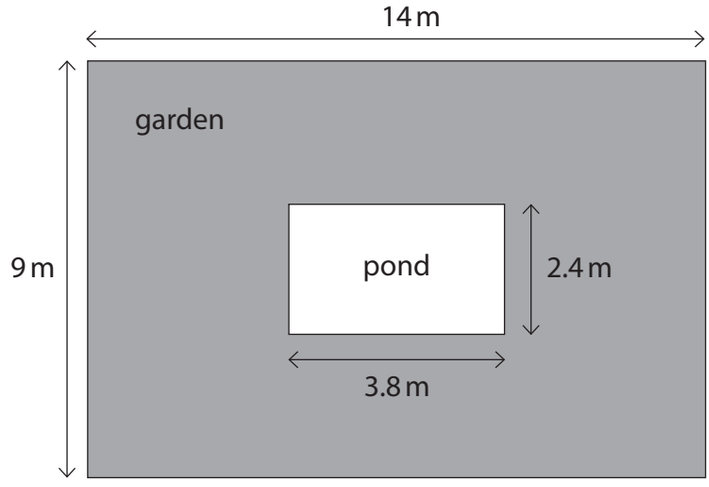
Conclusion: No, £1200 is not enough! [A1]

Alternative: Cost per chair inc. VAT = £73 × 1.20 = £87.60
14 × £87.60 = £1226.40 (> £1200 budget)

No *[A1]*

(Total for Question 2 is 4 marks)

3 Here is a garden and a pond.
The garden and the pond are rectangular.



Kayla is covering the garden with grass seed.
She will not put any grass seed in the pond.

One box of grass seed covers 15 m^2 of garden.

Work out the number of boxes of grass seed Kayla needs to cover the garden.

(5)

Step 1: Calculate total area of the garden (14 m by 9 m):

Garden area = 14 × 9 = 126 m^2 [M1]

Step 2: Calculate area of the rectangular pond (3.8 m by 2.4 m):

Pond area = 3.8 × 2.4 = 9.12 m^2 [M1]

Step 3: Calculate area to be seeded (garden minus pond):

Grass area = 126 − 9.12 = 116.88 m^2 [M1]

(Continued on next page to calculate seed boxes -->)

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Step 4: Calculate number of boxes of grass seed required:

One box covers 15 m^2

Number of boxes = 116.88 ÷ 15 = 7.792 boxes [M1]

Step 5: Round up to the next whole box:

- Grass seed can only be purchased in whole boxes.
- 7 boxes cover: 7 × 15 = 105 m^2 (leaves 11.88 m^2 unseeded)
- **Therefore, Kayla must buy 8 boxes [A1]**

Verification check:

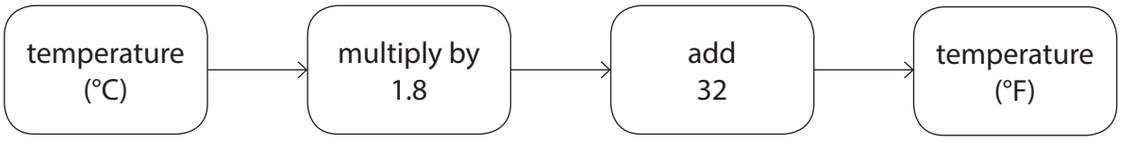
8 boxes × 15 m^2 = 120 m^2 (120 m^2 > 116.88 m^2, fully covers garden)

8 [A1]

(Total for Question 3 is 5 marks)

4 Dave is going on holiday to Madrid.
The temperature tomorrow in Madrid will be 21°C.

Dave uses this rule to change the temperature into Fahrenheit (°F).



Dave thinks the temperature in Madrid will be more than 70°F.

Is Dave correct?
Show why you think this.

(3)

Step 1: Multiply temperature in °C by 1.8:

21 × 1.8 = 37.8 [M1]

Step 2: Add 32 to convert to Fahrenheit (°F):

37.8 + 32 = 69.8 °F [M1]

Step 3: Compare with Dave's belief (> 70 °F):

- *Calculated temperature = 69.8 °F*
- *69.8 °F < 70 °F (it is slightly under 70 °F)*

Conclusion: No, Dave is not correct! [A1]

Alternative reverse method:

- *(70 - 32) ÷ 1.8 = 38 ÷ 1.8 ≈ 21.11 °C*
- *21.11 °C > 21 °C (Madrid is cooler than 70 °F)*

No *[A1]*

(Total for Question 4 is 3 marks)

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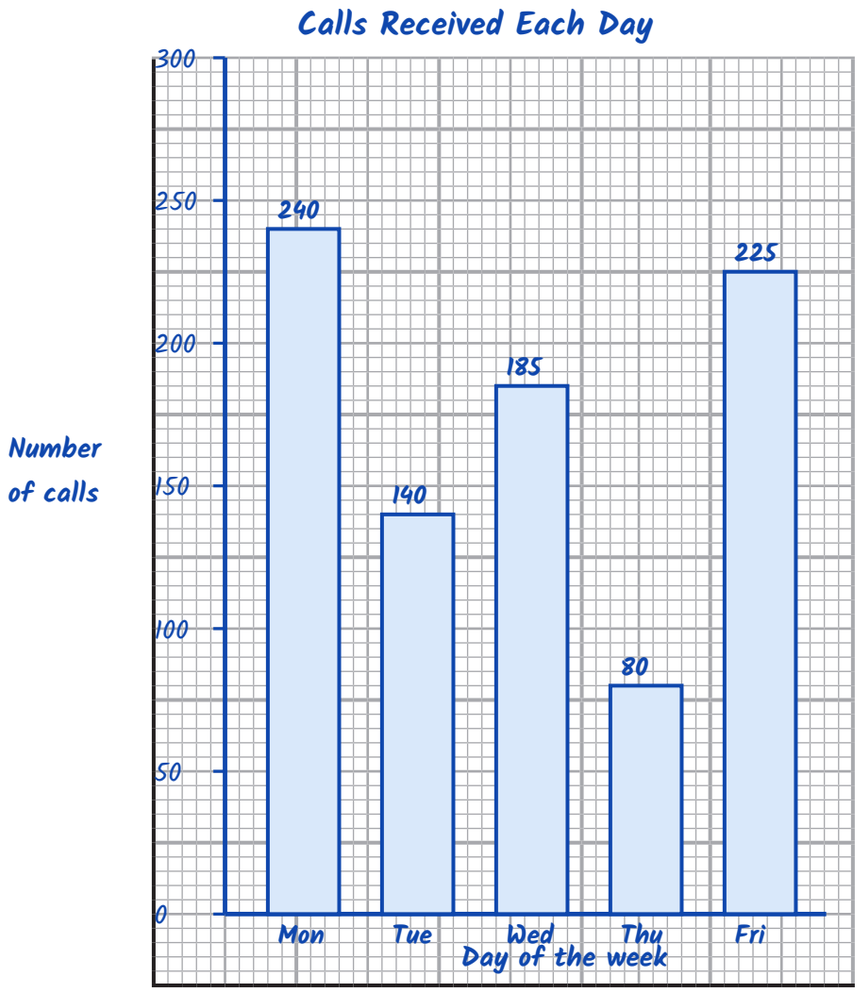
5 Kai works in a call centre.

The table shows the number of phone calls he receives each day.

Monday	Tuesday	Wednesday	Thursday	Friday
240	140	185	80	225

Draw a bar chart to display this information.

(3)



Linear vertical scale with clear labels [A1]

Accurate bar widths & plotting of all 5 days [B1, C1]

(Total for Question 5 is 3 marks)

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6 Seenal is a baker.
She buys flour in 50 kg bags.

The table shows the number of 50 kg bags of flour she used to make loaves in each of the last 4 weeks.

week	1	2	3	4
number of bags of flour	7	14	8	13

Seenal will make 2400 loaves in week 5
Each of these loaves will need 250 g of flour.

Seenal works out the weekly mean number of 50 kg bags of flour used in these 5 weeks.
She will use this figure for future orders.

(a) Work out the weekly mean number of 50 kg bags of flour used in these 5 weeks.

(5)

Step 1: Convert flour weight per loaf into kilograms:

250 g = 250 ÷ 1000 = 0.25 kg per loaf [A1]

Step 2: Calculate total flour needed for 2400 loaves in week 5:

2400 × 0.25 kg = 600 kg of flour [B1]

Step 3: Calculate number of 50 kg bags of flour used in week 5:

600 kg ÷ 50 kg = 12 bags of flour [B1]

Step 4: Sum the number of bags across all 5 weeks:

Total bags = 7 + 14 + 8 + 13 + 12 = 54 bags [C1]

(Continued on next page to calculate mean and check -->)

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Step 5: Work out the weekly mean number of 50 kg bags:

Mean = Total bags ÷ Number of weeks

Mean = 54 ÷ 5 = 10.8 bags [D1]

· *Or 11 bags (rounded to nearest whole bag) [E1]*

10.8 *[E1]*

☑ (b) Show a check of your calculation for the mean.

(1)

Reverse calculation check of the mean calculation:

10.8 × 5 = 54 bags (matches total across 5 weeks) [F1]

· *Or: 11 × 5 = 55 bags (close to 54)*

(Total for Question 6 is 6 marks)

7 Gordon buys a new slide for his garden.

His garden measures 6 m by 7 m.

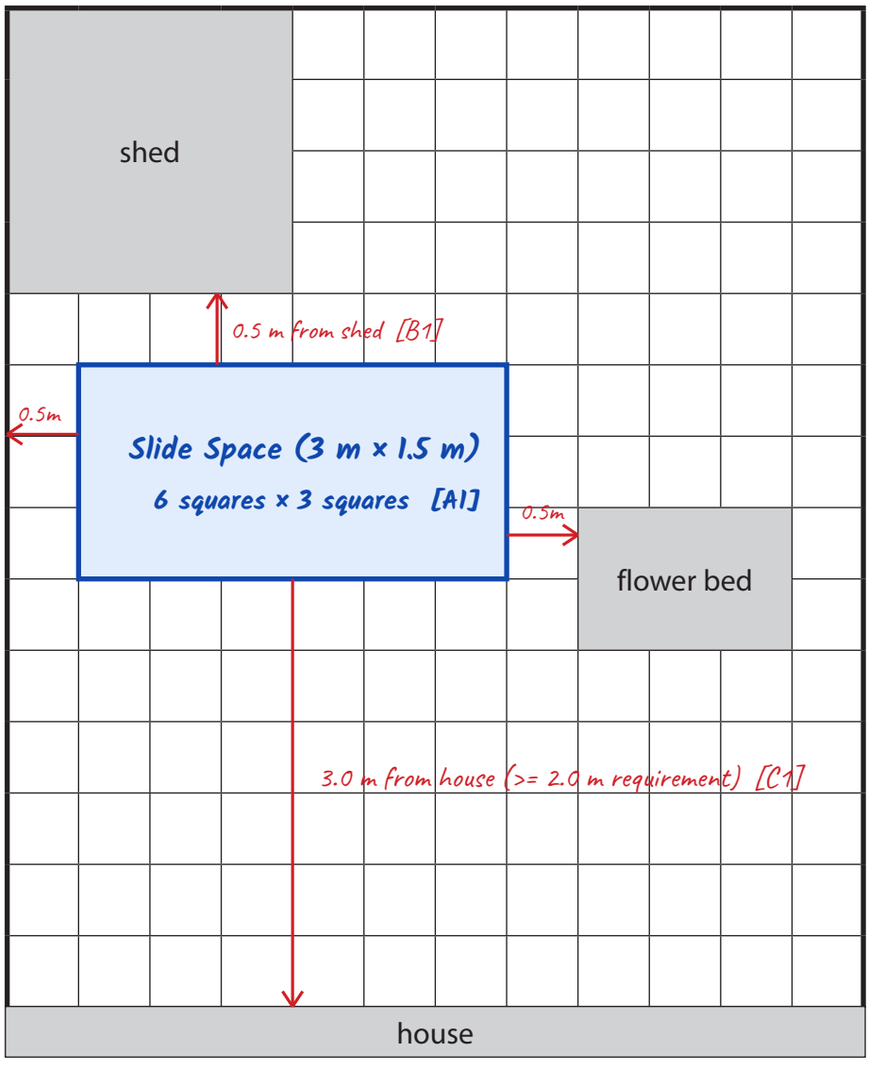
The space for the slide must be
- a rectangle, 3 m by 1.5 m
- at least 2 m away from the house
- at least 0.5 m away from a fence or any other feature.

Draw a space for the slide on the square grid below.

(3)

Key 1 cm on the grid represents 50 cm in the garden

—— fence



(Total for Question 7 is 3 marks)

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8 Belinda asks 20 people

'How many hours of TV did you watch last week?'

Here are the results.

3	17	4	4	6
11	14	14	1	20
9	8	9	6	12
7	8	13	13	9

Belinda wants to show these results in a frequency table.
She will use 4 equal groups.
The first group will start with 1 hour and the last group will end with 20 hours.

Design and complete a suitable frequency table.

(3)

Step 1: Divide range 1 to 20 into 4 equal group intervals of 5 hours:

Groups: 1 to 5, 6 to 10, 11 to 15, 16 to 20 [A1]

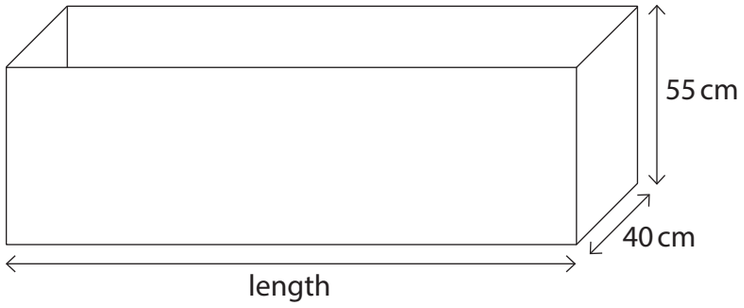
Hours of TV	Tally	Frequency
1 to 5	////	4
6 to 10	//// ///	8
11 to 15	//// /	6
16 to 20	//	2
Total		20

Clear table structure with appropriate headings [B1]

All 20 data points accurately tallied into equal frequency groups [C1]

(Total for Question 8 is 3 marks)

9 Dennis is making a container for tomato plants.
The container will be in the shape of a cuboid.



The capacity of the container will be 180 litres.

1 litre = 1000 cm^3

Work out the length of the container.
Give your answer to the nearest whole centimetre.

(4)

Step 1: Convert capacity from litres to cm^3:

180 litres = 180 × 1000 = 180 000 cm^3 [A1]

Step 2: Calculate cross-sectional area (width × height):

Area = 40 cm × 55 cm = 2200 cm^2 [B1]

Step 3: Calculate length of the container (Volume ÷ Area):

Length = 180 000 ÷ 2200 = 1800 ÷ 22 [M1]

= 81.8181... cm

Step 4: Round answer to the nearest whole centimetre:

• 81.8181... cm rounds up to 82 cm [D1]

Verification check: 82 × 40 × 55 = 180 400 cm^3 ≈ 180 litres

82 *[D1]* cm

(Total for Question 9 is 4 marks)

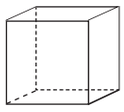
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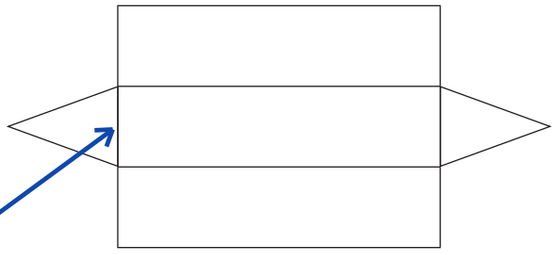
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10 (a) Match each 3-D shape to the correct net.
Draw lines to show your answers.

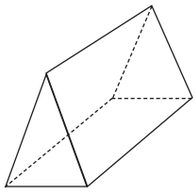
[A1, B1] **(2)**

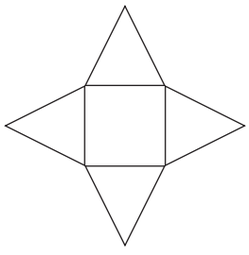
Cube (6 square faces)



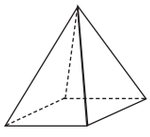


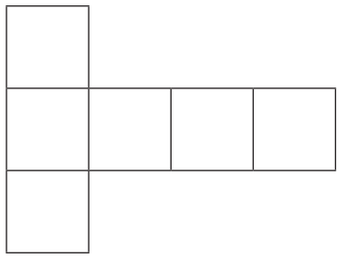
Triangular prism (5 faces)



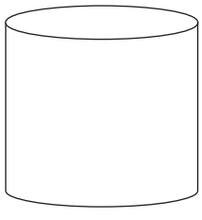


Square-based pyramid (5 faces)





Here is a cylinder.



Plan view = looking directly from above.

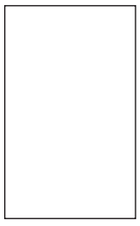
A cylinder viewed from above is a circle.

(b) Which one of the shapes below could be the plan view of this cylinder?
Tick the correct shape.

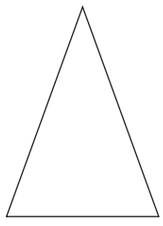
(1)



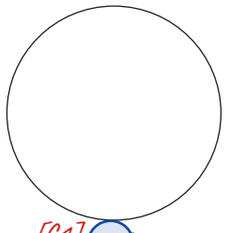
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[C1] ✓

(Total for Question 10 is 3 marks)

11 James wants to buy a house.
The price of the house is a quarter of a million pounds.

James has to pay a deposit for the house.
The deposit is 15% of the price of the house.

James already has part of the deposit.
He has £32 000

(a) How much **more** money does James need for the deposit?

(4)

Step 1: Write house price in standard digits:

A quarter of a million pounds = £250 000 [A1]

Step 2: Calculate 15% deposit on the £250 000 house price:

Deposit = 0.15 × £250 000 = £37 500 [M1]

- 10% of £250 000 = £25 000
- 5% of £250 000 = £12 500
- 15% = £25 000 + £12 500 = £37 500

Step 3: Calculate remaining money James needs for the deposit:

James already has £32 000.

Extra money needed = £37 500 – £32 000
= £5500 [M1, A1]

(Continued on next page for answer and percentage check -->)

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£ 5500 [A1]

(b) Show a check of your percentage calculation.

(1)

Reverse percentage check of calculation:

£37 500 ÷ 15 × 100 = £2500 × 100 = £250 000 [E1]

(Matches original quarter of a million pounds price)

(Total for Question 11 is 5 marks)

TOTAL FOR SECTION B IS 42 MARKS
TOTAL FOR PAPER IS 56 MARKS

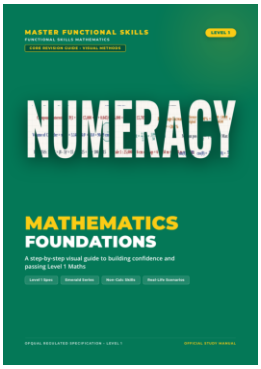
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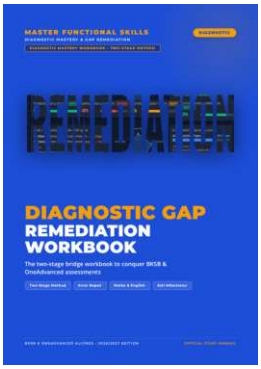


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